

IEOR 4101: Probability, Statistics and Simulation

Lecture 9: Large-Sample Properties

Large-Sample Behavior

- We want to investigate the behavior of “averages” over independent and identically distribution (i.i.d.) random variables
- Given i.i.d. r.v.'s $X_1, \dots, X_n$, consider the *sample mean*

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

- Question: when n is large, sample mean $\approx$ true mean?
- More questions:
 - What does it mean to be a good approximation?
 - How “large” is large?

Large-Sample Behavior

Motivations:

- **Monte Carlo simulation** (next topic): we generate random variates to approximate some difficult dynamics
- **Statistics** (next next topic): we use data to approximate unknown model parameters

In both cases, it is important to analyze the accuracy of the approximation

The theorems we discuss form the foundation of simulation and statistical analysis

Sample Mean

- Given n i.i.d. data $X_1, \dots, X_n$, we like to estimate $\mu = E[X]$

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

$$E[\bar{X}] = E\left[\frac{X_1 + \dots + X_n}{n}\right] = \frac{1}{n} (E(X_1) + \dots + E(X_n))$$

$$= \frac{1}{n} (n\mu) = \mu$$

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{1}{n^2} (\text{Var}(X_1) + \dots + \text{Var}(X_n)) \\ &= \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n} \end{aligned}$$

Sample Mean

- Given n i.i.d. data $X_1, \dots, X_n$, we like to estimate $\mu = E[X]$
- Mean of $\bar{X}$ is μ :

$$E[\bar{X}] = E\left[\frac{X_1 + \dots + X_n}{n}\right] = \mu$$

Sample Mean

- Given n i.i.d. data $X_1, \dots, X_n$, we like to estimate $\mu = E[X]$
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$$E[\bar{X}] = E\left[\frac{X_1 + \dots + X_n}{n}\right] = \mu$$

- Suppose $Var(X) = \sigma^2$. Variance of $\bar{X}$ is σ^2/n :

$$\begin{aligned} Var(\bar{X}) &= Var\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{Var(X_1 + \dots + X_n)}{n^2} = \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

We know $E(\bar{X}) = \mu$

This doesn't mean $\bar{X} \approx \mu$

We also know $\text{var}(\bar{X}) = \frac{\sigma^2}{n}$

So when n is large, $\text{var}(\bar{X}) \approx 0$

This suggests that $\bar{X} \approx \mu$ when n is large.

Law of Large Numbers

When n is large, the sample mean

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

is close to the true mean μ

Theorem (Law of Large Numbers). Let $X_1, X_2, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then, as $n \rightarrow \infty$, $P(|\bar{X} - \mu| \text{ is small}) \rightarrow 1$.

When n is large, $\bar{X} \approx \mu$ with probability
tending to 1.

Markov's Inequality

Theorem (Markov's Inequality). Let X be a nonnegative random variable with finite mean $E[X]$. Then, for any $a > 0$,

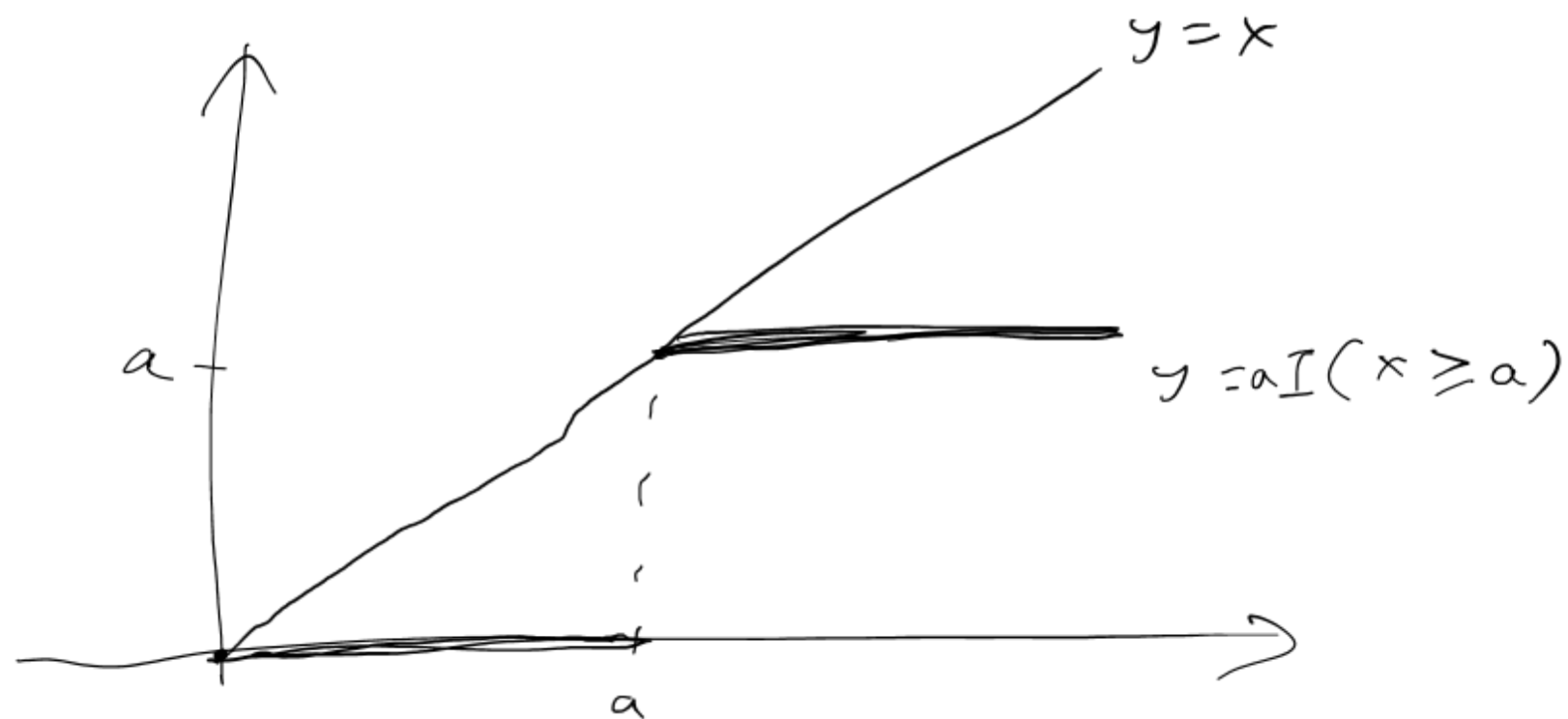
$$P(X \geq a) \leq E[X]/a$$

Proof.

For any $a > 0$, $\mathbf{1}\{X \geq a\} \leq X/a$, where $\mathbf{1}\{X \geq a\} = 1$ if $X \geq a$ and 0 otherwise

So $E[\mathbf{1}\{X \geq a\}] \leq E[X/a]$

or equivalently $P(X \geq a) \leq E[X]/a$



$$a I(x \geq a) \leq x$$

$$a I(X \geq a) \leq X$$

viewing X as a random variable,
take expectation on both sides

$$E[a I(X \geq a)] \leq E[X]$$

$$a E[I(X \geq a)] \leq E[X]$$

$$a P(X \geq a) \leq E[X]$$

Chebyshev's Inequality

Theorem (Chebyshev's Inequality). Let X be a random variable with finite mean μ and finite variance σ^2 . Then for any $k > 0$,

$$P(|X - \mu| \geq k) \leq \sigma^2/k^2$$

Proof. Apply Markov's inequality to the random variable $Y^2 = |X - \mu|^2$

Then $P(|X - \mu| \geq k) = P(Y \geq k) = P(Y^2 \geq k^2) \leq E[Y^2]/k^2 = \sigma^2/k^2$

Take $|X - \mu|^2$ as the random variable
and a as k^2 in Markov inequality

$$P(|X - \mu|^2 \geq k^2) \leq \frac{E|X - \mu|^2}{k^2}$$

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Law of Large Numbers

Theorem (Law of Large Numbers) [more precise version]. Let $X_1, X_2, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then, as $n \rightarrow \infty$,

$$P(|\bar{X} - \mu| \leq \epsilon) \rightarrow 1$$

for any $\epsilon > 0$.

Proof.

Apply Chebyshev's inequality to $\bar{X}$:

Mean of $\bar{X}$ is μ

Variance of $\bar{X}$ is σ^2/n

So, for any $\epsilon > 0$,

$$P(|\bar{X} - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Equivalently,

$$P(|\bar{X} - \mu| \leq \epsilon) \rightarrow 1$$

Take $\bar{X}$ as the random variable in Chebyshev inequality.

$$E[\bar{X}] = \mu, \quad \text{var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$P(|\bar{X} - \mu| \geq k) \leq \frac{\text{var}(\bar{X})}{k^2} = \frac{\sigma^2}{k^2 n} \rightarrow 0$$

as $n \rightarrow \infty$.

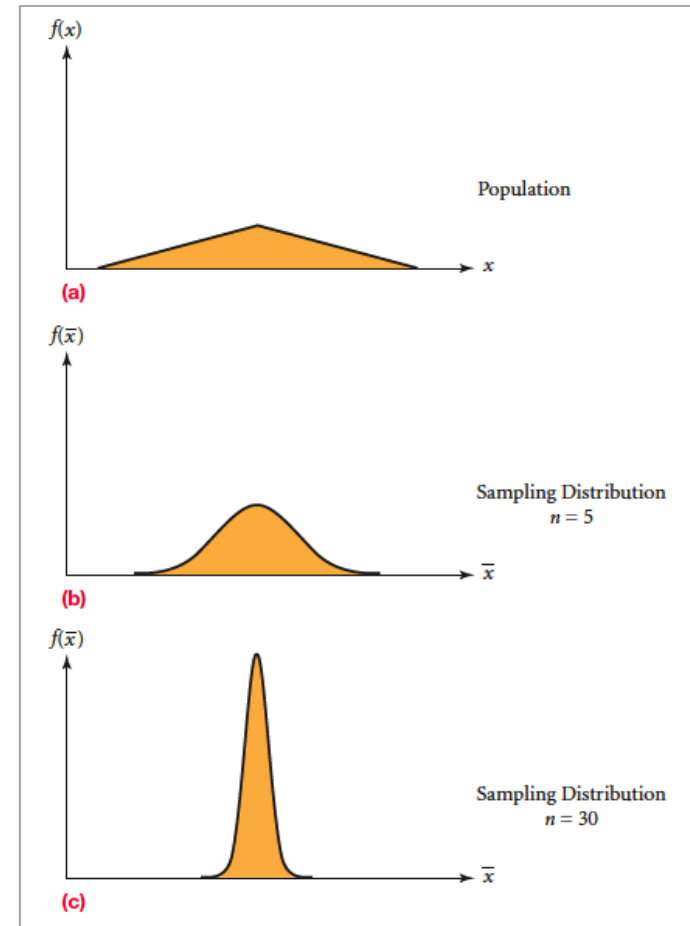
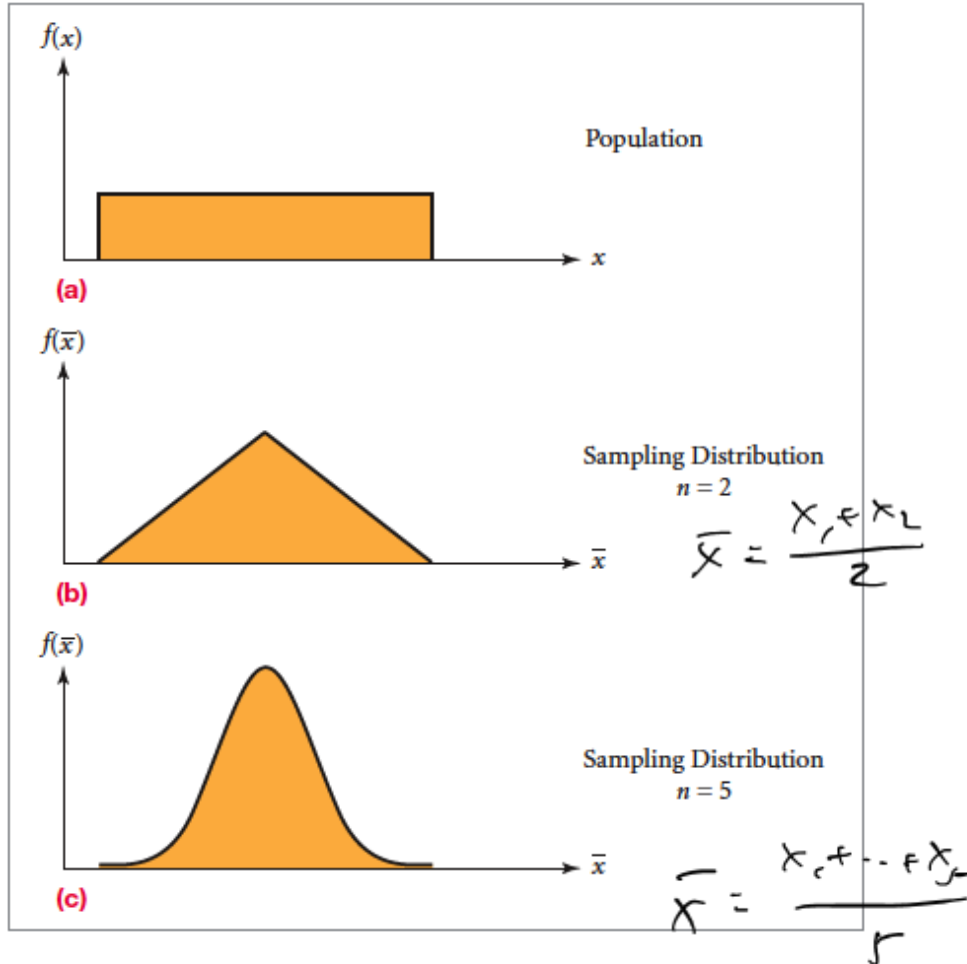
Equivalently, $P(|\bar{X} - \mu| < k) \rightarrow 1$ as $n \rightarrow \infty$

Central Limit Theorem

- From the law of large numbers, we know that the sample mean is a good estimate of the true expectation, *provided* that the sample size is large
- How large is large enough?
- **Theorem (Central Limit Theorem).** Let $X_1, X_2, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then, for large n , the distribution of $\bar{X}$ is approximately normal with mean μ and variance σ^2/n .

$$\bar{X} \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

Central Limit Theorem



Central Limit Theorem

- Implications of the central limit theorem:

$$P(\bar{X} \leq x) \approx P(Y \leq x)$$

where $Y \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

- The approximation can also be written as

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq x\right) \approx P(Z \leq x)$$

where $Z \sim N(0,1)$

- A rule of thumb is that the above approximation holds when n is at least 30

Sample vs Sampling Distributions

- The distribution of $\bar{X}$ is called the *sampling distribution*
- The object $\bar{X}$ (often called a statistic; will be visited later) and its distribution depends on n i.i.d. data $X_1, \dots, X_n$
- This distribution relates to, but not the same as, the distribution of any sample point X_i , e.g.,
 - mean of X_i = mean of $\bar{X}$ = μ
 - var of X_i = σ^2 , but var of $\bar{X}$ = σ^2/n
- By the law of large numbers, we expect the sampling distribution of $\bar{X}$ to concentrate around μ
- By the central limit theorem, we know the sampling distribution of $\bar{X}$ is approximately a normal distribution with mean μ and variance σ^2/n

Example

The weights of a population of workers have mean 167 and standard deviation 27. If a sample of 36 workers is chosen, approximate the probability that the sample mean of their weights lies between 163 and 170.

$$\bar{X} \overset{\text{approx.}}{\sim} N\left(167, \frac{27^2}{36}\right)$$

$$P(163 < \bar{X} < 170)$$

$$= P\left(\frac{163 - 167}{27/\sqrt{36}} < \frac{\bar{X} - 167}{27/\sqrt{36}} < \frac{170 - 167}{27/\sqrt{36}}\right)$$

$$= P\left(\frac{163 - 167}{27/\sqrt{36}} < Z < \frac{170 - 167}{27/\sqrt{36}}\right) \text{ where } Z \sim N(0, 1)$$

$$= P\left(\frac{170-167}{27/\sqrt{36}}\right) - P\left(\frac{167-167}{27/\sqrt{36}}\right)$$

⋮

Example

An insurance company has 25,000 automobile policy holders. If the yearly claim of a policy holder is a random variable with mean 320 and standard deviation 540, approximate the probability that the total yearly claim exceeds 8.3 million.

$$\text{Generally, } \bar{X} \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\sum_{i=1}^n X_i = n\bar{X} \overset{\text{approx.}}{\sim} N(n\mu, n\sigma^2)$$

In this problem,

$$\begin{array}{l} \text{total} \\ \text{yearly} \\ \text{claim} \end{array} = \sum_{i=1}^{25000} X_i \overset{\text{approx}}{\sim} N(25000 \times 320, 25000 \times 540^2)$$

$$P(\text{total yearly claim} > \text{£.3 million})$$

$$= P\left(\sum_{i=1}^{25000} X_i > \text{£.3 million}\right)$$

$$= P\left(\frac{\sum_{i=1}^{25000} X_i - 25000 \times 320}{\sqrt{25000 \times 540^2}} > \frac{\text{£.3 million} - 25000 \times 320}{\sqrt{25000 \times 540^2}}\right)$$

$$= P\left(Z > \frac{\text{£.3 million} - 25000 \times 320}{\sqrt{25000 \times 540^2}}\right) \text{ where } Z \sim N(0, 1)$$

Example

The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

$$\text{Let } X = \text{\# students attending the college} \\ \sim \text{Bin}(450, 0.3)$$

$$\text{or } X = \sum_{i=1}^{450} X_i \quad \text{where} \quad X_i = \begin{cases} 1 & \text{if admitted student } i \text{ attends,} \\ 0 & \text{otherwise} \end{cases}$$

S.

$$X = \sum_{i=1}^{450} x_i \sim \text{approx } N(450 \times 0.3, 450 \times 0.3 \times (1-0.3))$$

$$P(X > 150) = P\left(Z > \frac{150 - 450 \times 0.3}{\sqrt{450 \times 0.3 \times (1-0.3)}}\right)$$

where $Z \sim N(0, 1)$

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